AI-Based Semantic Similarity and Novelty Assessment System for Academic Project Proposals

Alternative technical title:
An Intelligent NLP-Based System for Semantic Similarity Detection and Novelty Assessment of Academic Projects

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2. Domain

- Artificial Intelligence and Machine Learning
- Natural Language Processing (NLP)
- Information Retrieval
- Semantic Search
- Web Application Development
- Educational Technology
- Explainable AI

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3. Introduction

Every academic year, colleges receive a large number of project proposals from UG and PG students. These proposals generally contain a project title, abstract, problem statement, objectives, methodology, datasets, technologies, and expected outcomes.

A major challenge for project coordinators and departmental committees is determining whether a newly submitted project is **similar to or repetitive of an existing project** completed in previous academic years or currently being developed by another student group.

The problem becomes more difficult when two projects use different terminology but address essentially the same problem.

For example:

Existing project:
"CNN-Based Plant Disease Identification"

New project:
"Deep Learning Approach for Automatic Detection of Crop Leaf Diseases"

Although the wording is different, both projects may have a highly similar problem, application domain, and methodology.

Traditional keyword or string matching may fail to identify such semantic similarity.

Therefore, this project proposes an **AI-based Academic Project Similarity and Novelty Assessment System** that maintains a centralized repository of previous and ongoing academic projects and uses NLP and semantic embedding techniques to compare newly submitted proposals against them.

The system will not simply identify matching words. It will analyze the **semantic relationship between projects**, identify potentially repeated project ideas, highlight common components, identify differences, and provide an explainable similarity report to the project coordinator.

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4. Problem Statement

Problem Statement

Colleges maintain a large number of UG and PG project proposals across departments and academic years. Currently, checking whether a newly proposed project is similar to a previous or ongoing project is largely a manual process.

Project coordinators need to examine titles, abstracts, problem statements, objectives, methodologies, datasets, and technologies across potentially hundreds or thousands of projects. This process is time-consuming, subjective, and difficult to perform consistently across departments.

Moreover, projects with different wording can address the same underlying problem, making conventional keyword-based comparison insufficient.

Hence, there is a need for an **intelligent centralized system that automatically compares new academic project proposals with previous and ongoing projects using NLP and semantic similarity techniques, identifies potentially repetitive projects, explains the sources of similarity and difference, and assists coordinators in making project approval decisions.**

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5. Aim

To develop an **AI-powered web-based system that automatically analyzes and compares academic project proposals with previous and ongoing college projects using NLP-based semantic similarity techniques and provides an explainable similarity and novelty assessment to assist project coordinators in project screening and approval.**

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6. Objectives

Primary Objectives

1. Develop a centralized repository for storing UG and PG academic project information.
2. Provide a web-based interface through which student groups can submit project proposals.
3. Analyse project titles, abstracts, problem statements, objectives, methodologies, datasets, and technologies.
4. Apply NLP preprocessing techniques to normalize and prepare project information.
5. Calculate **lexical similarity** using traditional text similarity techniques.
6. Calculate **semantic similarity** using transformer-based sentence/document embeddings.
7. Compare newly submitted projects against:
    - Previous-year projects
    - Current/ongoing projects
    - Projects from the same department
    - Projects from other departments
    - UG and PG projects
8. Generate an overall similarity score using a suitable hybrid similarity model.
9. Identify the specific project components responsible for high similarity.
10. Identify differences between the proposed and existing projects.
11. Develop a **novelty indicator** based on differences in methodology, dataset, application domain, technology, and objectives.
12. Generate an explainable similarity report for project coordinators.
13. Provide configurable similarity thresholds, such as:

- Less than 10% → Low similarity / eligible for preliminary approval
- 10–25% → Moderate similarity / review recommended
- 25–50% → High similarity / revision or committee review
- Above 50% → Very high similarity / strong possibility of repetition

14. Keep the final approval decision under the control of the project coordinator/committee rather than making the AI system the sole decision-maker.
15. Evaluate the performance of different similarity techniques and identify the most suitable approach for academic project screening.